10.3.4 The implicit Q theorem

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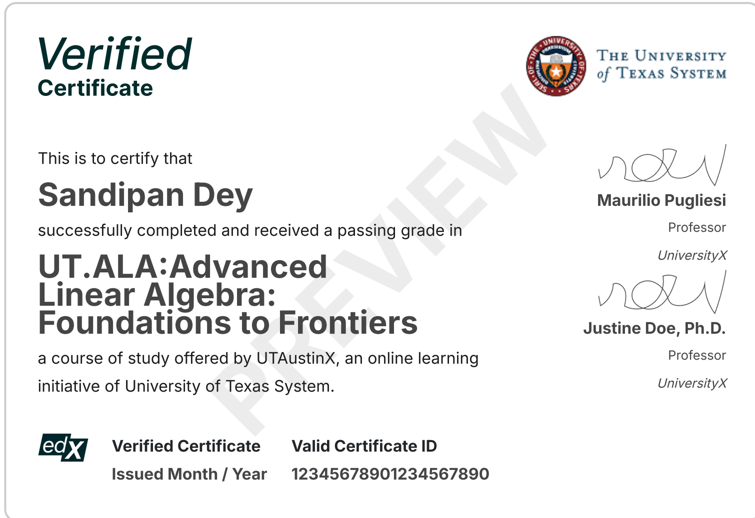

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plus beta_1 0

times q_1. Okay? So these two must be equal. Now A is known, q_0 is known, q_0 is

known here. Hmm, we know that q_0 is orthogonal to q_1. So just like when we

came up with the algorithm for computing the Gram-Schmidt process, we say, "Oh, we

can then compute beta_0 0 by saying let's look at q_0 transpose A q_0." That

must be equal to beta_0 0.

So at this point we've

6:55 / 6:55

2.0x

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Homework 10.3.4.1

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Give all the details of the proof of Theorem 10.3.4.2 in the notes .

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Solution:

For a complete solution, see the notes .

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Answers are displayed within the problem

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- if we actually tried to form with you then we would

have to take our matrix G_0 and hit it from the right with the next Givens

rotation. But remember that Givens rotation when applied from the right

only affects the second and third column. Importantly it leaves the first column

alone. So you get a bunch of non-zeros here. As a matter of fact, you get a non-

zero here, here, and here, typically. Okay? Then you go to the

3:30 / 3:30

2.0x

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